Socialnetwork Diffusion with LLMs

An LLM agent-based model over the BCDJ microfinance networks

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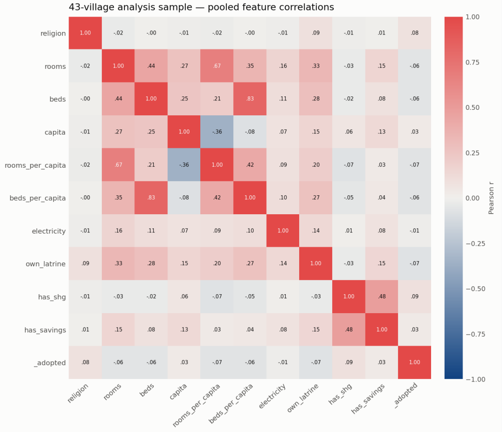
Data interpretation

Modelling choices

- Agents are households; the individual survey covers a median 46.8% of a village.
- Covariates kept as narrative grounding, not as predictors: wealth proxies, financial exposure (SHG, savings), in/out-group standing (subcaste, religion).
- `leader` is implied in adoption, explicit in transmission – it is what the seeding routine conditions on; `adopted` likewise becomes conditional there.
- Edge type dropped: nothing records who told whom, so there is no ground truth for which of the 12 relations carried influence. We run on the union network.
- Also dropped: redundant, high-missingness, redacted, near-constant and bare-identifier fields.



n=9598 households across 43 villages. capita / has_shg / has_savings / *_per_capita limited to the individual-survey subset; religion is nominal (category codes, not a magnitude); pairwise-complete correlations.

Two decision-theoretic games

Transmission $G^T = \langle A, S, P_i, U_i \rangle$

Will this household inform the neighbour?

- $A = \{\text{YES}, \ \text{NO}\}$
- $S = \{\text{THEY JOIN}, \ \text{THEY DON'T JOIN}\}$
- U_i: subjective risk/reward; P_i: beliefs over S

Played by every `informed` household, each turn, for each not-yet-informed / not-yet-adopted neighbour $j \in N$.

Assumption: transmission is not a neutral act. Consequences resemble a coordination game — *if I joined and told them, and they join too, I gain*.

Wider scope: an action space with intent — ENDORSE / INFORM / OPPOSE / IGNORE.

Adoption $G^A = \langle A, S, P_i, U_i \rangle$

Will this household adopt microfinance?

- $A = \{\text{YES}, \ \text{NO}\}$
- $S = \{\text{BENEFICIAL}, \ \text{LIMITED}, \ \text{HARMFUL}\}$
- U_i: subjective risk/reward; P_i: beliefs over S

Played once, whenever a household becomes `informed`.

S is the time-agnostic coarsening of what a loan does to a household's finances: positive (house improvement), neutral (balance transfer), negative (misuse, debt spiral).

BCDJ factorise diffusion into the same two conditionally independent decisions — a 7-parameter logit and a 2-parameter Bernoulli. We substitute one factor at a time.

Agent architecture: a structured wrapper, not off-the-shelf agency

1. The loop I would build — OODA

- Observe — neighbour states, what was said, village context
- Orient — household profile, priors, in/out-group standing
- Decide — the game above, over an explicit utility structure
- Act — adopt, or tell a specific neighbour

Models cognitive *and* physical features and carries environmental uncertainty explicitly. Strongly structured architectures (old-school Wooldridge) wrapping LLMs, rather than assuming inherent agency.

2. The loop actually implemented

- Receive — one transmission is delivered
- Decide — one LLM call, one decision
- Advance — state update, next round

Loosely OODA-shaped, but sequential and stateless. Full OODA was cut for time and complexity, not because it is the wrong model.

[OODA loop figure]

Simplifying assumptions taken from the original authors to save runtime: agents are informed once, decide once, and process one transmission at a time.

What varies: three axes

1. Information set

What the agent is told.

- Own profile (A)
- Neighbour's profile (B)
- Endorsement (C) / leader (L)

Each at three levels of richness: absent, listed facts, or an LLM-written narrative fixed per household id.

2. Instruction design

How the agent is asked to reason.

- Plain (D0)
- Model of appropriateness (D1)
- Decision-theoretic (D2)

Drawn from cognitive theory and decision theory — not prompt engineering folklore.

3. LLM choice

Who is answering.

- `gpt-4.1-nano`, `gpt-5-nano`
- `gpt-5.4-nano` (primary)
- `gpt-5.6-luna`
- `grok-4.20`, `grok-4.1-fast`

Non-reasoning models, so hidden processes do not confound the prompt-design effect.

`gpt-5.4-nano` is the primary agent: cheapest recent SOTA API model, fast, and in game-theoretic work smaller models tend to behave more human-like. Claude is absent only because it does not accept the same strict JSON schema through the OpenAI-compatible API.

The prompt design grid: A B C/L D

Axis	**0**	**1**	**2**
A — ego profile	no household info	data as a listing	LLM-written narrative
B — neighbour profile	no neighbour info	data as a listing	LLM-written narrative
C — endorsement *(adoption)*	decision not mentioned	neighbour's decision given	—
L — leader *(transmission)*	not a seed	seeded by the MFI	—
D — response format	plain	model of appropriateness	decision-theoretic

D1 — model of appropriateness. The agent answers, in order:

1. What kind of situation is this?
2. What kind of person am I?
3. What should someone like me do in a situation like this?

D2 — decision-theoretic. The agent must return a strict JSON schema populating the decision matrix of G^A/G^T — P_i over S, U_i over $A \times S$ — with a justification per cell, then choose.

This is what makes the reasoning auditable: the decision can be checked against the agent's *own* matrix.

Adoption: $3 \times 3 \times 2 \times 3 = 54$ designs. Transmission: the same grid with C replaced by the conditional L.

Three experiments, increasing autonomy

Pilot

Which prompt design is worth paying for?

- No simulation loop — two hand-built contrastive samples from village 6: one real adopter, one real non-adopter.
- Full factorial, both games, $n = 25$ reps/arm.
- Fisher's exact per design, FDR-corrected.
- Plus: do D2 designs obey their own matrix?

Hybrid

Does an LLM beat a fitted logit at take-up?

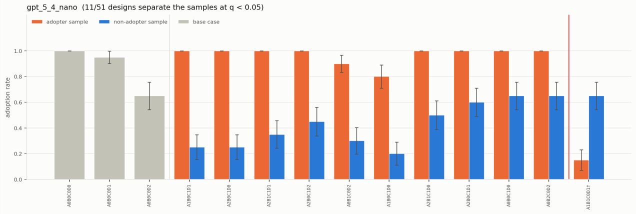
- BCDJ's Bernoulli transmission at q_N, q_P, transliterated unchanged.
- Only adoption is an LLM call — that single substitution is the whole experiment, which is what licenses attributing any gap to it.
- Mirror arm (LLM transmission, algorithmic adoption): `NotImplementedError`.

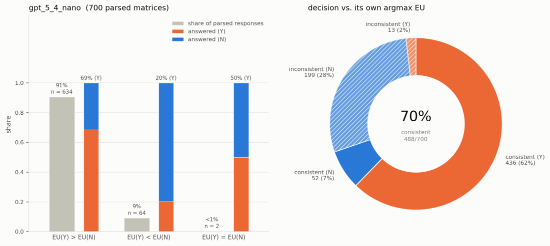
Full LLM

Can an ABM with zero fitted parameters reach a 2-parameter structural model?

- Adoption and per-edge transmission are both LLM calls.
- Seeded only at the empirical leaders; BCDJ's timing kept.
- Scored against real take-up *and* BCDJ's logit on the same network.
- Six models, two villages.

Pilot I — adoption





11 of 51 designs separate a real adopter from a real non-adopter at $q < 0.05$; one inverts (right of the red line).

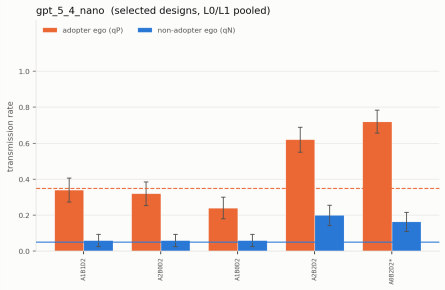
Only two survivors are non-endorsed — the agents pick up far more from the *neighbour's decision* than from the demographic data.[a]

The D2 audit. 700 parsed matrices; 91% put $EU(Y) > EU(N)$, yet the decision matches the agent's own argmax only 70% of the time — and the 28% that break it break it by answering *no* to their own *yes*.

Forcing reasoning through a semi-formal bottleneck buys legibility, and immediately exposes the inconsistency it made visible.

[a] On this sample pair. Needs an in-depth analysis before it becomes a claim.

Pilot II — transmission

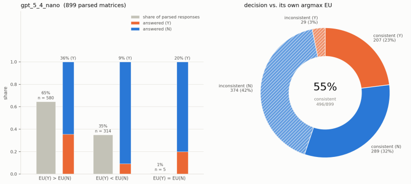


Top 5 lowest-rate designs, L0/L1 pooled. Dashed/solid lines are the paper's fitted $q_P = 0.35$ and $q_N = 0.05$.

No ground truth here — transmission is observed nowhere in the bundle. BCDJ never observed it either: q_N, q_P are identified from the adoption pattern with informed status integrated out. So the comparison is to the paper's fitted rates, and the test is face validity only.

Lean designs land close: `A1B0D2` gives 0.24/0.06 against the paper's 0.35/0.05, preserving the taker/non-taker asymmetry. Richer profiles (`A2B2D2`, `A0B2D2`) inflate both rates and wash the asymmetry out.

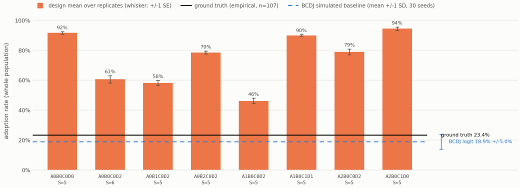
The D2 audit is worse here: 899 parsed matrices, only 55% consistent with their own argmax — 42% say *no* where their own matrix says *yes*.[a]



[a] A gap our logical-enhancement frameworks are built to close.

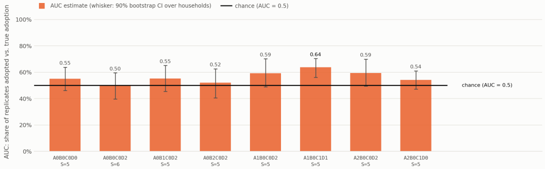
Hybrid — incomplete, and instructive

Village 6 -- adoption rate by design, against the take-up it is aiming at



denominator: full giant-component village, n=107 (a household never asked was never informed and counts as not adopted). every design overshoots the village's real take-up, by 2.0x to 4.0x. the baseline's spread is +/-1 SD over its seeds, not the SE of its mean: it stands in for one realisation of village 6, which is what a design is compared against.

Village 6 -- does a design rank households correctly, even at the wrong level?



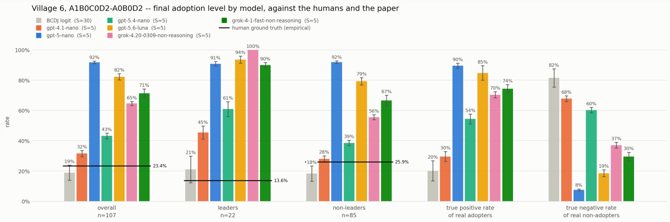
1 of 8 designs keep their whole interval above chance (those estimates are printed in black). the CI is 500 bootstrap resamples over households, not over replicates: it says how far the AUC estimate would move under a different draw of village 6's households, not how many more replicates the design needs. a design at S replicates can only score a household at multiples of 1/S, so a small S coarsens the ranking it is being credited with.

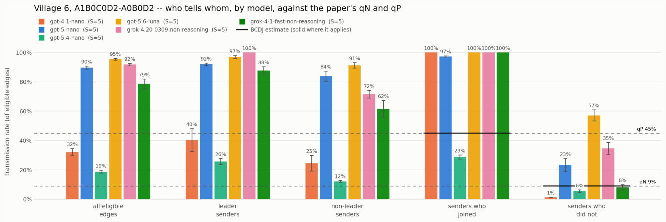
The failure, plainly. All 8 designs overshoot village 6's real take-up by 2.0–4.0× (23.4% → 46–94%); BCDJ's logit sits at 18.9%. Nor is it a ranking problem behind a level problem: AUC is 0.50–0.64, only 1 of 8 keeps its interval above chance, and 8 of 8 put leaders above non-leaders where the truth is 13.6% against 25.9%.

Where the divergence may come from

- The pilot scored a *contrast* between two curated samples; the loop demands an absolute *level*, and nothing calibrates it.
- The pilot separated mostly through endorsement (C); under BCDJ's Bernoulli flood almost everyone is informed, so C fires everywhere and stops discriminating.
- Mechanism transfer is not free: context-sampling-to-LLM and LLM-to-LLM behave differently from algorithmic-transmission-to-LLM.
- The mirror arm was never run, so the ablation that would localise this is missing.

Full LLM — village 6 (A1B0C0D2–A0B0D2)





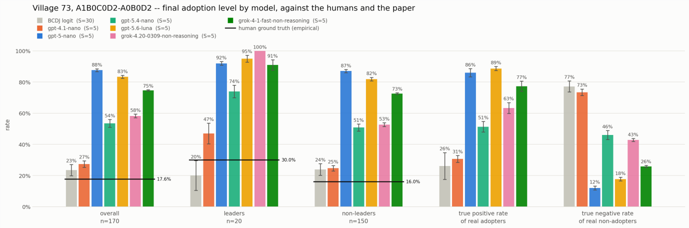
One design pair, six models, $n = 107$ households.

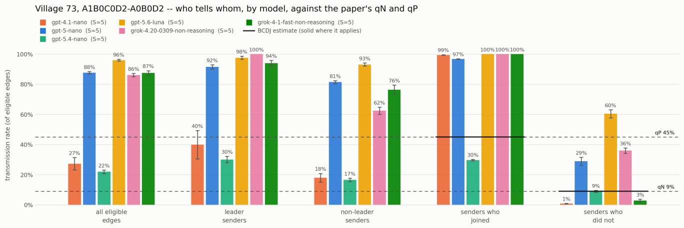
Smaller is closer. `gpt-4.1-nano` lands at 32% against a 23.4% truth and beats the BCDJ logit on true-positive rate. Newer `gpt-5-nano` / `gpt-5.6-luna` hit 82–92%, and their true negative rate collapses to 8–19%.

Everyone gets leaders backwards. Truth: leaders 13.6%, non-leaders 25.9%. Every model — and BCDJ's own logit — puts leaders on top.

Transmission. `gpt-5.4-nano` is the only model near the paper's rates (19% vs q_P 45% / q_N 9%) and the only one keeping the taker asymmetry. The rest fire on 79–95% of eligible edges: everyone tells everyone, and the network stops mattering.

Full LLM — village 73, out of sample





Same configuration, denser network, $n = 170$. Nothing was re-tuned.

Why this village. Village 73 flips the leader gap: leaders adopt at 30.0% against non-leaders' 16.0%, the opposite of the average case. A second axis for asking whether responses are grounded in the data at all.

The honest reading. Every model puts leaders above non-leaders here too — so they now agree with the truth, but for the *same* reason they contradicted it in village 6. This is a constant prior on status, not sensitivity to the village.

The level story transfers unchanged: `gpt-4.1-nano` at 27% against a 17.6% truth, everything newer at 54–88%.

What could an LLM actually buy us here?

1. Mechanism, not just fit

A calibrated, verified agent produces a reasoning trace that is a *plausible* account of why this household joined — the kind of explanation a human would give in the same circumstances. Two fitted numbers, q_N and q_P, cannot say anything about why.

2. Latent heterogeneity the covariates do not carry

The best single covariate explains $R^2 = 0.009$ of adoption. If social context, in/out-group standing and who-told-you matter, an LLM is one of the few ways to enter them into the model at all — and it can generate counterfactual populations, not merely re-fit the observed one.

3. Portability without a fitted parameter

The full-LLM model has zero fitted parameters: it ran on village 73 untouched. A structural model must be re-estimated per setting — that is the trade this study is pricing.

Not yet demonstrated. Current results show sensitivity to social context but no advantage over the simpler model: levels overshoot 2–4 $\times$, household ranking is near chance, the leader effect is a prior rather than a reading of the data, and D2 traces contradict their own matrices 30–45% of the time. Stronger claims need reasoning ground truth.

Stretch target: a Western comparison population

The structure this study needs is demographics + network + observed adoption. For a WEIRD population that combination is essentially unavailable off the shelf — but experimental economics has close cousins, where the demographics are collected and "adoption" is a recorded choice under a known incentive structure.

Candidate paradigms

- **Public goods games** — contribution and conditional cooperation; the nearest analogue to take-up under social influence.
- **Sequential / multi-stage games** — trust, centipede, bargaining: the same decision-under-a-signal shape as G^A.
- **Market entry games** — coordination with a congestion externality, close to G^T.
- **Signalling games** — for the endorsement channel (C) that drove the pilot separation.

What transfers, what does not

- Transfers: the agent architecture, the prompt-design grid, the D2 self-consistency audit, the whole scoring apparatus.
- Does not: the real social network. Lab populations are anonymous and randomly rematched — the topology would have to be commissioned, or recovered from a field experiment.

Centaur (Binz et al., 2024) is the proof the direction is tractable: one model fine-tuned across 160 experiments, predicting held-out human behaviour including in *unseen* paradigms. The natural benchmark — and a natural source of the behavioural ground truth this study lacks.

Thank you